

## 1. 2015Q34

Questions 34 and 35 refer to the list of factors shown below.

- (1) Smoking
- (2) Family history
- (3) Overweight
- (4) Radiation

Which of the above factors can be controlled by lifestyle adjustment?

- A. (1) and (3) only
- B. (2) and (4) only
- C. (1), (2) and (3) only
- D. (1), (3) and (4) only

## 2. 2015Q35

Which of the above are risk factors for coronary heart disease?

- A. (1) and (3) only
- B. (1), (2) and (3) only
- C. (1), (2) and (4) only
- D. (2), (3) and (4) only

## LQ

## 1. 2014Q9

Table I shows the number of deaths in Hong Kong caused by certain diseases. The deaths are categorized according to sex and age.

**Table I**

| Cause of death    | Sex    | Age group |   |     |      |       |       |      |
|-------------------|--------|-----------|---|-----|------|-------|-------|------|
|                   |        | All ages  | 0 | 0-4 | 5-14 | 15-44 | 45-64 | ≥65  |
| Heart disease     | Male   | 3352      | 1 | 2   | 2    | 104   | 679   | 2564 |
|                   | Female | 2981      | 2 | 2   | 0    | 20    | 146   | 2811 |
| Diabetes mellitus | Male   | 213       | 0 | 0   | 0    | 2     | 43    | 168  |
|                   | Female | 246       | 0 | 0   | 0    | 10    | 19    | 217  |
| Colon cancer      | Male   | 725       | 0 | 0   | 0    | 16    | 177   | 532  |
|                   | Female | 627       | 0 | 0   | 0    | 8     | 131   | 488  |

- a) Rank the diseases in the descending order of mortality (death rate). (1 mark)
- b) From the data above, generalize a trend of mortality that is exhibited by all the diseases. (1 mark)
- c) Table II shows the relative proportions of males and females in the above table who had particular lifestyles:

**Table II**

| Lifestyle                              | Male  | Female |
|----------------------------------------|-------|--------|
| Smoking daily                          | 78.9% | 21.1%  |
| Never smoked                           | 36.9% | 63.1%  |
| ≥4 servings of processed meat per week | 57.7% | 42.3%  |

For the following diseases, explain how these lifestyles of the males and females are related to the number of deaths in Table I.

Heart disease: (3 marks)

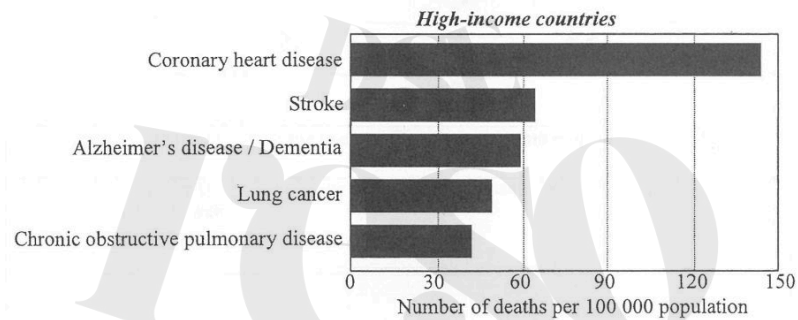
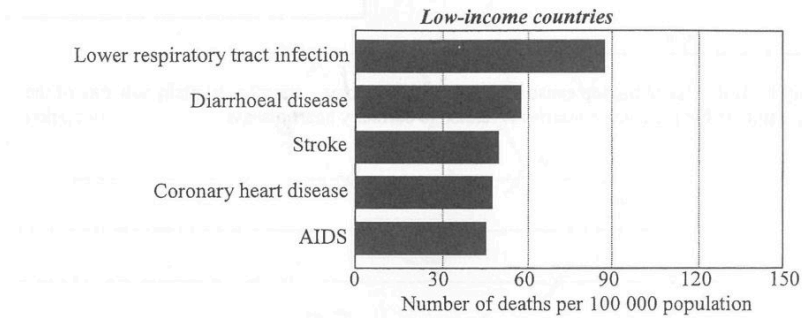
Colon cancer: (3 marks)

## 2. 2016Q6

- a) Kitty and Karen are identical twins. Kitty has preferred meats to vegetables in her diet since her childhood. Kitty suffered from colon cancer at age 35 while Karen had the same disease 10 years later.
  - i) Why did both sisters suffer from colon cancer? (1 mark)
  - ii) Why did the disease occur at different ages? (1 mark)
- b) Give **two** other lifestyles that increase the risk of suffering from cancer. (2 marks)

## 3. 2018Q4c

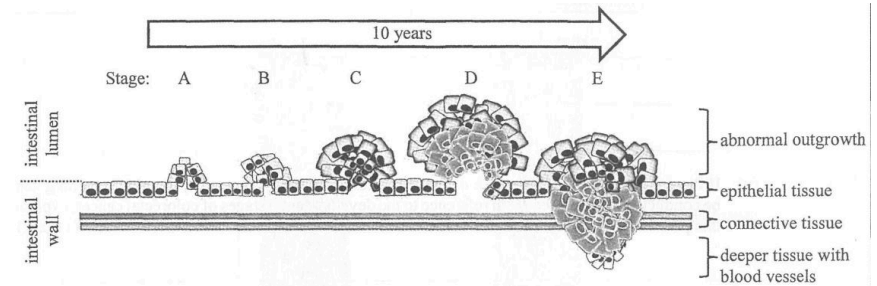
The bar charts below show the top five diseases that caused death in low-income countries and high income countries respectively in the year 2015:



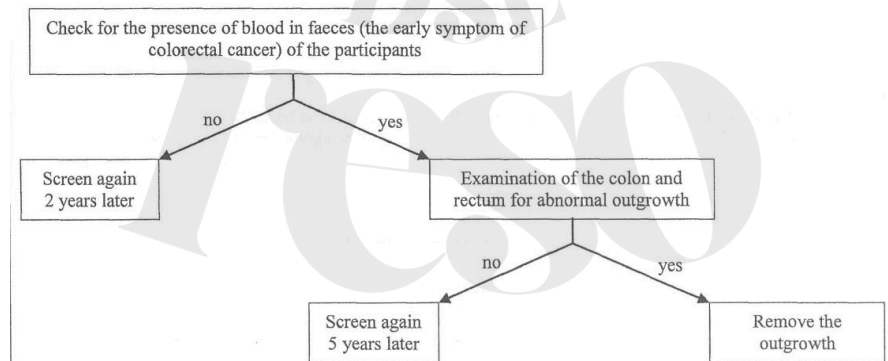
- c) Coronary heart disease is the top cause of death in high-income countries. Explain how **one** of the lifestyle habits in high-income countries is related to coronary heart disease. (4 marks)

## 4. 2022Q7

Colorectal cancer is one of the most common cancers in Hong Kong. The schematic diagram below shows the developmental stages of colorectal cancer over time:



- a) Which stage of colorectal cancer has a high risk of spreading? Explain your answer. (2 marks)
- b) The Department of Health in Hong Kong has launched a regular screening programme for the prevention of colorectal cancer. The flowchart below illustrates the screening programme:



- i) The screening programme is recommended for people aged 50 or above. Give **two** reasons why this group of people is more susceptible to colorectal cancer. (2 marks)
- ii) If there is no abnormal outgrowth of the epithelium of the large intestine, the next screening can be conducted 5 years later. With reference to the developmental stages of colorectal cancer, explain this practice. (1 mark)
- iii) Recently, there is a growing trend of people diagnosed with colorectal cancer at a younger age. Suggest **two** eating habits which may lead to this growing trend. (2 marks)

Ans

2015Q34: D (47%)

2015Q35: B (81%)

2014Q9

- 9 (a) heart diseases > colon cancer > diabetes mellitus (1) (1)
- (b) number of death increases with age groups (1) / age (1)
- (c) Heart disease:
- heart disease kills more male than females (1)
  - more male were (daily) smokers (1)
  - nicotine of cigarette smoke increases the chance of blocking the blood vessel (1) (3)
- Colon cancer:
- colon cancer kills more males than females (1)
  - more male consume processed meat (1)
  - processed food contains chemicals that may stimulate mutations / which are mutagens (1) of colon epithelium (3)

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8 marks

2016Q6

- 6 (a) (i) Karen is genetically identical to Kitty, hence they both have the cancer-causing genes (1) (1)
- (ii) Kitty's eating habit would trigger the development of colon cancer earlier (1) (1)
- (b) Any **two** of the following:
- smoking, excessive drinking, stress, lack of exercises (1,1) (2)

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4 marks

2018Q4c

- 8 (c) • consumption of high calorie food / fatty food / lack of exercise (1)
- increases the risk of plaque formation / deposition of cholesterol / fat in coronary arteries (1)
- which leads to narrowing of the lumen of arteries / blockage in arteries / decrease blood flow to the heart (1) (4)
- heart muscles do not have enough nutrients / food and oxygen supply (1), resulting in heart attack

2022Q7

- 7 (a) • E (1)
- the tumour cells can spread through the blood to other parts of the body (1) / the tumour cells have grown out of the boundary / break through the connective tissue (1) (2)
- (b) (i) 

|                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concept for mark award: <ul style="list-style-type: none"><li>chance of mutation in relation to time (1)</li><li>ability to repair in relation to age (1)</li></ul> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|

 (2)
- e.g.
- (c) • they have been living for a longer time period for the accumulation of mutations / they have been exposed to cancer-causing agents / carcinogens / mutagens for a longer period, therefore the chance for genetic mutation to occur is higher (1)
- the cells in old people have a lower ability to repair / old people have a weaker immune system / there is a lower level of expression of tumour suppressor genes / more active of oncogenes (1)
- (ii) • the development of the outgrowth to malignant tumour / the stage of spreading of cancer cells takes many years in colorectal cancer (1) (1)
- (iii) any **two** of the following:
- insufficient dietary fibre in their diet (1)
  - consumption of a lot of processed meat / processed food (1), which contains carcinogenic substances (2)
  - consumption of a lot of deep-fried food (1), which contains carcinogenic substances produced during deep-frying

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7 marks